An IoT-Based Online Monitoring System for Continuous Steel Casting

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***Abstract*—Monitoring solutions using the Internet of Things (IoT) techniques, can continuously gather sensory data, such as temperature and pressure, and provide abundant information for a monitoring center. Nevertheless, the heterogeneous and massive data bring significant challenges to real-time monitoring and decision making, particularly in time-sensitive industrial environments. This paper presents an online monitoring system based on an IoT system architecture which is composed of four layers: 1) sensing; 2) network; 3) service resource; and 4) application layers. It integrates various data processing techniques including protocol conversion, data filtering, and data conversion. The proposed system has been implemented and demonstrated through a real continuous steel casting production line, and integrated with the TeamCenter platform. Results indicate that the proposed solution well addresses the challenge of heterogeneous data and multiple communication protocols in real-world industrial environments.**

***Index Terms*—Continuous steel casting, data processing, Internet of Things (IoT), online monitoring.**

I. Introduction

An ONLINE monitoring system is essential to monitor environment conditions and working status of major equipment (e.g., continuous steel casting equipment) in industrial fields, so as to achieve accurate failure prediction, timely repair, and maintenance of the equipment. Such system contributes to the integration of operation, maintenance, and management of the equipment. Nevertheless, being limited by the severe conditions of industrial environments, most measuring methods are unreliable or too expensive to maintain. Continuous steel casting is a crucial transition process between steelmaking and rolling in steel production industry, representing a typical harsh industrial environment. Compared with ingot casting, continuous steel casting has a number of advantages, including improvement of steel quality and yield, and savings of energy and manpower. Currently, continuous steel casting accounts for approximately 95% of steel production worldwide [1]. With the enormously growing significance of the continuous casting products, the demand for online monitoring and maintenance, repair, and overhaul (MRO) [2] services has been constantly rising.

The Internet of Things (IoT) [3] provides a promising solution for online monitoring and related activities, along with wireless sensor networks (WSNs) [4] and mobile Internet [5]. IoT has developed from a convergence of technologies including wireless communication [6], the Internet, and microelectromechanical systems [7]. IoT has a close relationship with machine-to-machine (M2M) [8] communication in smart manufacturing, since M2M provides a comprehensive solution with data acquisition, data transmission, data analytics, and service management. Technically speaking, a typical IoT system architecture is composed of three layers: 1) sensing; 2) network; and 3) application layers. The foundation of IoT is networks of massive sensors or mobile terminals that sense and gather data constantly [9]. In IoT, WSN is in charge of gathering and transferring environment data [10]. An IoT-based online monitoring system is composed of a large number of wireless sensor nodes, which are deployed to an industrial file to collect data such as temperature, pressure, and speed. Various types of sensors and data communication protocols generate a heterogeneous and sophisticated IoT environment. The sensed data are transferred in raw or processed form to a server via a base station (or sink node), where incoming data can be further analyzed. As a result, faults and some other related events can be detected at the server without manual and human-centric operations. However, a number of issues need to be addressed when applying IoT techniques in online monitoring.

Based on a detailed investigation of continuous steel casting environments and a comprehensive review of the related research literature, this paper proposes an IoT-based online monitoring system for continuous steel casting. It extends the traditional three-layer IoT system architecture to four layers: 1) sensing; 2) network; 3) service resource; and 4) application layers. It integrates various data processing techniques including protocol conversion, data filtering, and data conversion to deal with the large volume of dynamic, geographically distributed and heterogeneous field data. The proposed solution is validated through a real-world case study in industrial settings.

The remainder of this paper is organized as follows. Section II provides a literature review. Section III presents the proposed framework and data processing methods. Section IV illustrates a real IoT-based application case study to validate

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the proposed approach. Section V concludes this paper and discusses our future work.

II. Literature Review

To date, the industrial IoT has been emerging as a novel technical pattern enabling a widespread application of the IoT in industry where physical sensors gather data from the industrial field and deliver the message to the Internet. The collected real-time big data, in turn, contribute to the online monitoring of manufacturing environment and the optimizing of entire industry [11].

However, there are serious challenges since industrial applications have strict requirements on delay, reliability, and security. A number of research efforts have been made to cope with the above challenges. Kong *et al.* [12] proposed an environmental space time improved compressive sensing algorithm for data loss and reconstruction in sensor networks, based on the features of spatial correlation, temporal stability, and low-rank structure. To deal with the event reporting scenario, Pan *et al.* [13] designed an event data collection mechanism for ZigBee tree-based WSNs, including a tree formation algorithm, slot assignment schemes, and event mode operations. Lu *et al.* [14] proposed a security solution for smart grid communications and information integration, which adopted digital watermarking to achieve the secure communications. Yan *et al.* [15] developed two secure data transmission protocols for cluster-based WSNs by using the identity-based digital signature scheme and the identity-based online/offline digital signature scheme, separately. Ishmanov *et al.* [16] proposed a lightweight and robust trust establishment scheme to address the attack-resistance and robustness issues in WSNs. Kumar *et al.* [17] introduced an adaptive mutual authentication framework with the characteristics of symmetric key cryptography considered, so as to support mutual authentication, key establishment, message confidentiality, and node identity and location privacy. Yu *et al.* [18] presented a secure routing protocol to detect internal attacks such as a Byzantine attack using both message and route redundancy during route discovery. Islam *et al.* [19] provided a comprehensive review of the reliability and security challenges of WSNs in factory automation environments.

On the other hand, Cloud computing and other distributed computing techniques have been explored to deal with the massive data. Tao *et al.* [20] investigated the applications of IoT technologies in Cloud manufacturing to obtain intelligent perception and access of various manufacturing resources. Lu *et al.* [21] summarized the current situation of electrical power monitoring system, and presented a Hadoop-based framework for power monitoring. Gonizzi *et al.* [22] presented a low-complexity distributed data replication method to increase the resilience and storage capacity of an IoT-based surveillance system. Due to the increasing needs in Big Data analytics such as data generated by the IoT, Liu *et al.* [23] conducted a detailed analysis on authenticator-based data integrity verification techniques on Cloud and IoT data.

Besides, Koedrith *et al.* [24] developed two image-based environmental monitoring sensor applications by utilizing low-power sensors and the Tenet general purpose sensing system. Aslan *et al.* [25] proposed a framework for forest fire detection and monitoring by use of WSNs. They explored the WSN architecture, sensor deployment scheme, and clustering and communication protocols. A WSN test-bed was developed by Zhou *et al.* [26] to monitor the indoor climate of a large-scale space. Although these efforts on communication protocols and data processing are not addressing issues specifically to monitoring applications, the approaches proposed are adaptable to various applications, including smart grid [27], environmental monitoring [24], fire detection [25], and smart building [28].

III. System Design and Data Processing Techniques

This section describes the proposed IoT-based framework design for continuous steel casting online monitoring systems. As shown in Fig. 1, the proposed system architecture has four layers: 1) sensing; 2) network; 3) service resource; and 4) application layers.

The sensing layer is at the bottom of the system architecture. The raw data are gathered at this layer, where various sensors, radio frequency identification (RFID) readers and tags, and other devices are deployed in industrial fields to sense, acquire and then record the data of machine status (parameters, such as temperature and pressure).

In the network layer, different transmission/reception protocols and techniques including ZigBee, CAN bus, RFID, and Bluetooth can be chosen on the basis of application requirements. In addition, communication protocols used by devices are transformed into a protocol by an intelligent gateway, which is a critical element for enabling the connectivity of industrial devices and other systems to the IoT. The gateway has the capability of integrating technologies and protocols for networking, embedded control, and industrial-grade security, which is used here to carry the Modbus/TCP protocol. Thus, a complex heterogeneous network is turned into a relatively simple homogeneous one.

The service resource layer has some basic functions, such as resource registration, service resource matching, resource state representation, and sensing device management. By searching the resource registration center, a service resource provider can be identified with the service request dynamically through certain matching algorithm. Sensing device management is in charge of coordinating and managing sensing devices. The service resource bus acts as a resource platform (B/S or C/S) to provide uniform resource locator and service interfaces for upper-layer applications.

The application layer offers an application container engine to pack, ship, and run any user-defined application, such as TeamCenter [29] and ERP [30]. As the Cloud computing technologies mature, application virtualization has been an irresistible trend. For example, Docker[1] is an open platform

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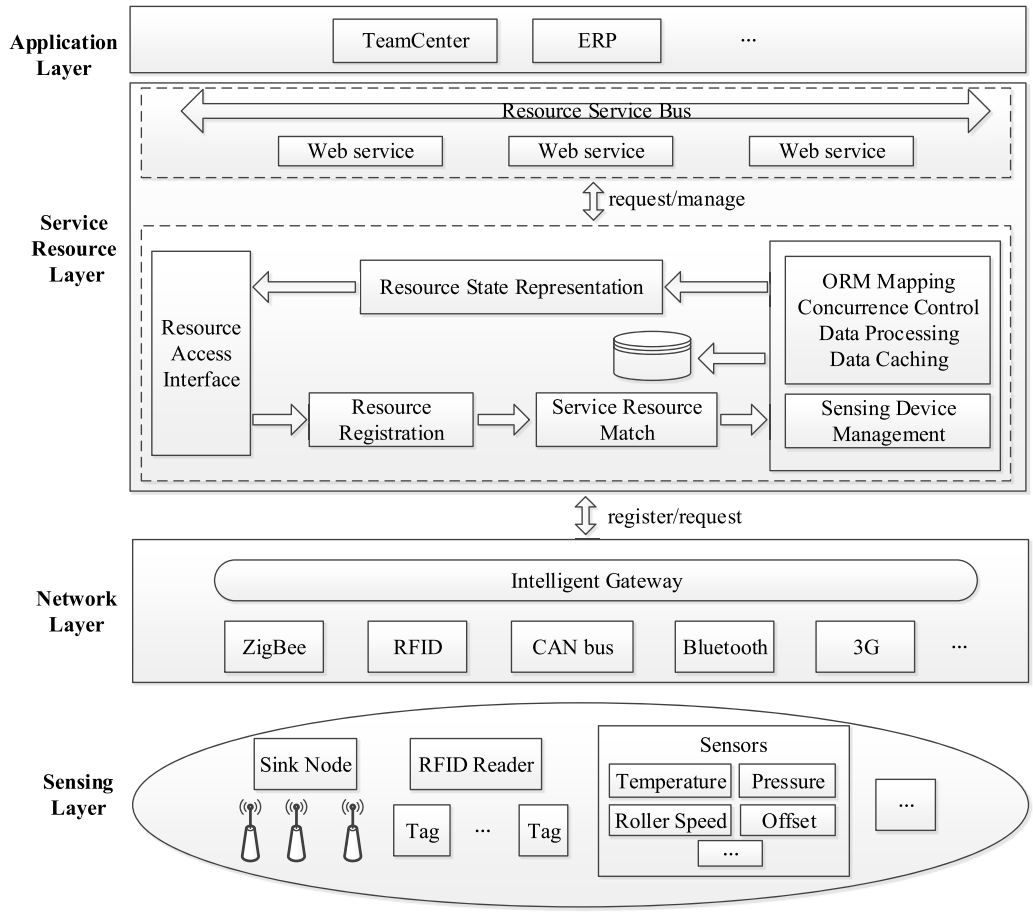

Fig. 1. Architecture of the online monitoring system for continuous steel casting.

for building, shipping, and running distributed applications, which are easy to modify, update, and scale independently.

A. Data Characteristics of the System

Real-time and reliable monitoring is essential in continuous casting steel plants. However, IoT-based solutions bring new challenges in achieving accurate monitoring and effective failure prediction for such equipment, due to the huge volume of dynamic, geographically distributed and heterogeneous data generated from the equipment.

1) Dynamic and Heterogeneous: The steel production environment is harsh, uncertain, heterogenic, and dynamic. Moreover, sensor nodes may move among different locations in some cases. Data gathered from various sensors in the device layer are transmitted by different communication protocols in the physical layer. Such dynamicity and heterogeneity result in the complexity of developing an appropriate data processing technique for the online monitoring system.

2) Volume and Velocity: The intrinsic nature is its high generation velocity, and a huge number of sensor nodes collecting data all the time. Thus, data stream usually arrives faster than we are able to deal with, and most conventional data processing techniques become insufficient. How to cope with the continuous, rapid, and changing data stream has urgently been a question to study for further data analysis and mining.

3) Uncertainty: Data from IoT-based systems have obvious uncertainty, including uncertainty in data themselves, semantic matching uncertainty, and analysis uncertainty. The data collected may change with the sensing environment. Moreover, data may also change during the transmission, which makes it crucial to remove the erroneous data and extract accurately the required information.

B. Data Processing Techniques

To address above mentioned challenges, this paper presents a set of data processing techniques to handle the data generated from the equipment based on the process of data stream.

1) Protocol Conversion: The data stream output of the physical layer involves diverse protocols, which increases the difficulty of operation, management, and maintenance for online monitoring. An intelligent gateway is utilized in communication network for a rapid processing of conversion between dissimilar protocols to transmit the data using a Modbus-TCP protocol, which is a variant of the Modbus family of simple communication protocols. The usage of intelligent gateway provides a data transmission medium for applications in the application layer to communicate with sensor nodes in the physical layer and the device layer via Modbus messaging using the TCP/IP protocols.

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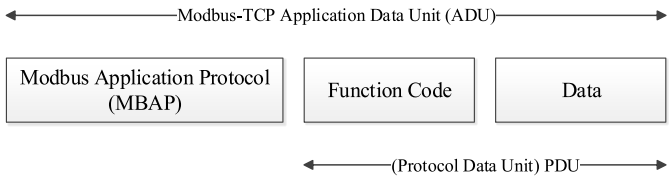

Fig. 2. Modbus/TCP data frame.

The Modbus-TCP protocol has become a standard protocol widely used in industrial networks. It is prescribed by the Modbus-TCP protocol that every sensor node or hardware device should be identified with a unique address so that a connection is easily recognized at the protocol level. Modbus-TCP data frame is, to some extent, a TCP frame with a standard Modbus data frame embedded, which is composed of three parts, i.e., data unit, function code, and packet header, as shown in Fig. 2. Modbus application protocol (MBAP) represents the header of data frame and is used to recognize the application data unit. A function code is used to define the requested action, any required data, and an error checking field.

2) Data Filtering: Data filtering in sensor nodes contributes to energy savings, and alleviates burdens of network bandwidth, computation, and storage. Therefore, data filtering is critically important to the communication network, which should be performed in sensor nodes before transmitting the data to the base station. Here, two kinds of filters are introduced.

a) Duplicated data filter: The data collected from sensors may contain a lot of duplicated data. The function of this filter is to filter them out. The gathered data could be described as a set of three properties

$$\text{NODE_DATA} = \{\text{Node_ID}, \text{Datavalue}, \text{Timestamp}\} \tag{1}$$

where NODE_DATA represents the data structure. Node_ID represents the unique identification of each sensor or device. Data value denotes the value measured by the sensor. Timestamp signifies the acquisition time of the data. During the process of data filtering, the data are stored in a hash table, which is taken as the keyword of Node_ID.

Assuming that a time interval T_{interval} is given, query the Node_ID in the hash table when new node data arrive. As shown in Fig. 3, if the Node_ID does not exist, the node data will be inserted into the hash table and output meanwhile; if the Node_ID exists already while the time interval between two collected node data is less than the value of T_{interval}, then the new node data are considered to be duplicated and will be discarded, update the Timestamp; if the Node_ID exists already but the time interval between two collected node data are greater than the value of T_{interval}, then the new node data are valid and allowed to be output, update the Data value and Timestamp in the hash table.

b) Event filter: The event filter selects the desired data that meet certain conditions. In an IoT application, what users are mostly interested in are the numbers of sensor nodes in a certain time and region, and the amount of active nodes.

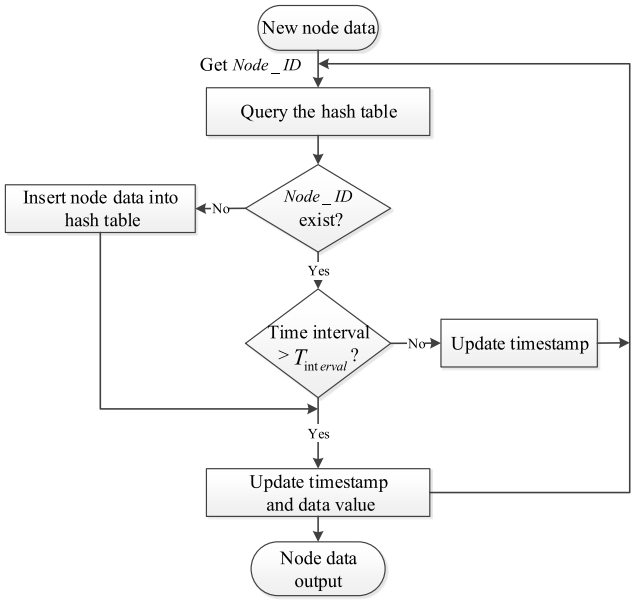

Fig. 3. Flowchart of duplicate data filter.

In this paper, the event filter is mainly used to manage three kinds of sensor nodes: 1) new nodes; 2) leaving nodes; and 3) currently active nodes. Q_{addnode} represents the queue that saves new sensor nodes, and $Q_{\text{deletenode}}$ represents the queue that saves dead nodes.

During a time interval $T_{\text{persisttime}}$, when a new node arrives, check the Node_ID to see whether the same Node_ID exits in the hash table. If it exits, check the interval of collection time. If the time interval is less than $T_{\text{persisttime}}$, update the Timestamp of the node data, choose this sensor node as an active one. If the time interval is greater than $T_{\text{persisttime}}$, delete the node data from the hash table and add it to $Q_{\text{deletenode}}$; we define this kind of sensor node as a dead node. If not, add the data to the hash table and add the node to queue Q_{addnode}.

3) Data Representation and Data Parsing: Extensible markup language (XML) is used to address the heterogeneous problem of data. The well-structured, scalable, self-descriptive features of XML render it to be independent of any applications. XML is utilized to represent and transmit the data in a unified form including status parameters of the equipment such as temperature, pressure, speed, and frame displacement, and control instructions sent from the application layer, as shown in Fig. 4. The XML representation contributes to further data parsing by means of document object model (DOM) or simple API for XML (SAX), and benefits the scalability and integration of data as well.

XML data parsing plays an important role in XML data processing since no operation can be done before an XML document is parsed. Data parsing involves parsing the MBAP header and the Modbus protocol data unit (PDU). Fig. 5 shows the flowchart of a data packet parsing strategy. The MBAP header contains a sensor address and part of the Modbus protocol. If the MBAP header contains the right address and

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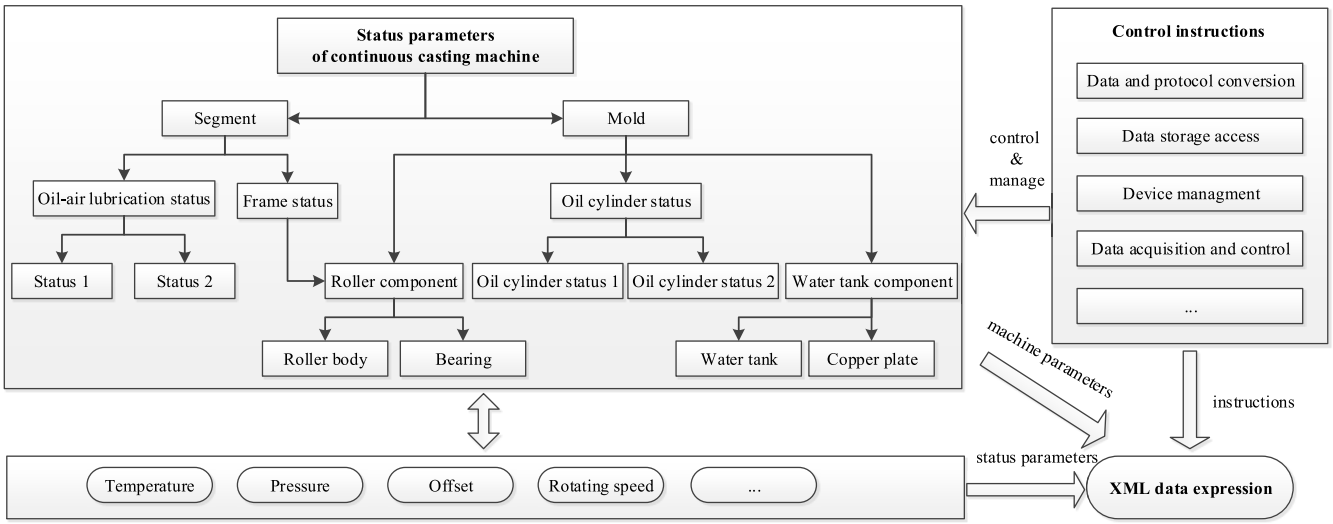

Fig. 4. XML data representation for continuous casting.

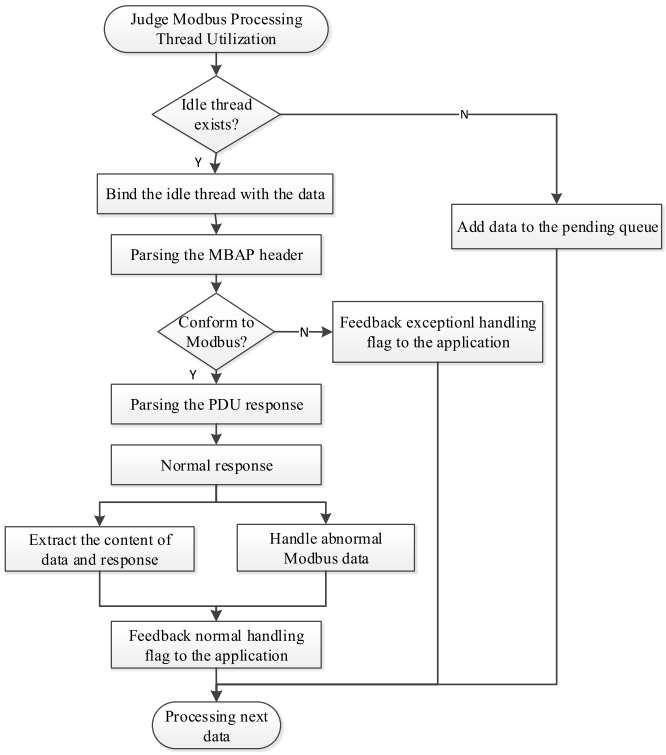

Fig. 5. Flowchart of data parsing strategy.

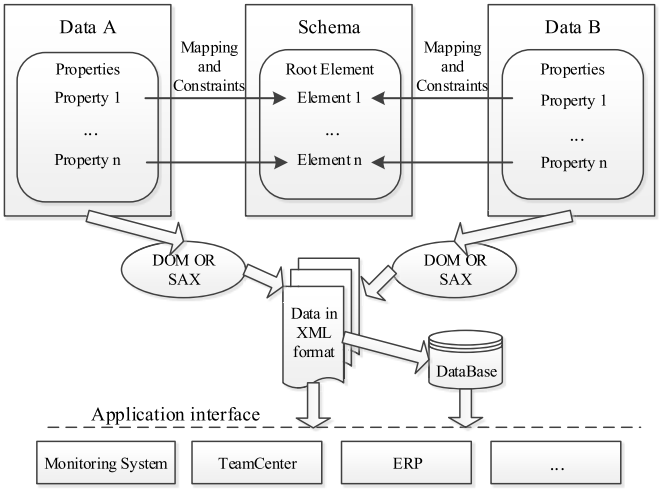

Fig. 6. Process of data validation.

Modbus protocol, it is considered as the right data in accordance with the Modbus protocol. If the data are Modbus protocol data, then continue to parse the PDU part of the data. If the PDU conforms to the normal response format, then extract the content of data and response. Otherwise, it is an abnormal Modbus protocol data item, and errors may appear in program or failures could occur in underlying sensors.

The aim of data parsing is to convert the input XML document and break it into small parts. SAX and DOM are two main approaches for XML data parsing. As an event-driven, serial-access mechanism, SAX reads an XML document as a stream and invokes callback functions provided by the application. Compared to SAX, DOM is much more complex. DOM represents an XML document as an object-oriented model, usually a tree structure. After the tree structure is established, applications are allowed to dynamically access and update their content and structure. In this paper, both approaches are used.

4) Data Validation: Due to the potential and unexpected change in the process of data parsing and exchange, data validation is required to filter incorrect and incorrectly formatted data. In this paper, Schema is used to establish the mapping and constraints between properties of data and elements of Schema. XML Schema is a language designed for describing the logical structure of XML document, in order to verify the validity of the logical structure.

As shown in Fig. 6, data A and data B stand for the collected data. The data are provided with properties including system ID, sensor location, value, and acquisition time. The role of DOM and SAX is to convert data, which are

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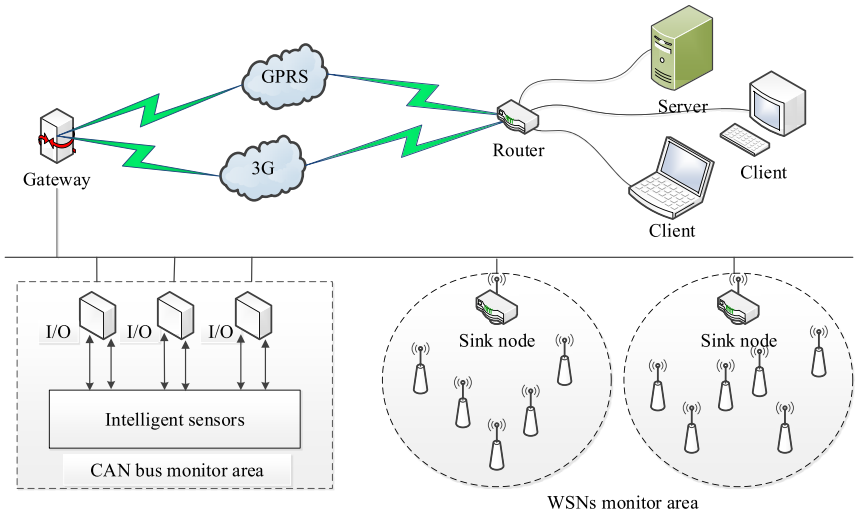

Fig. 7. Architecture of monitoring system.

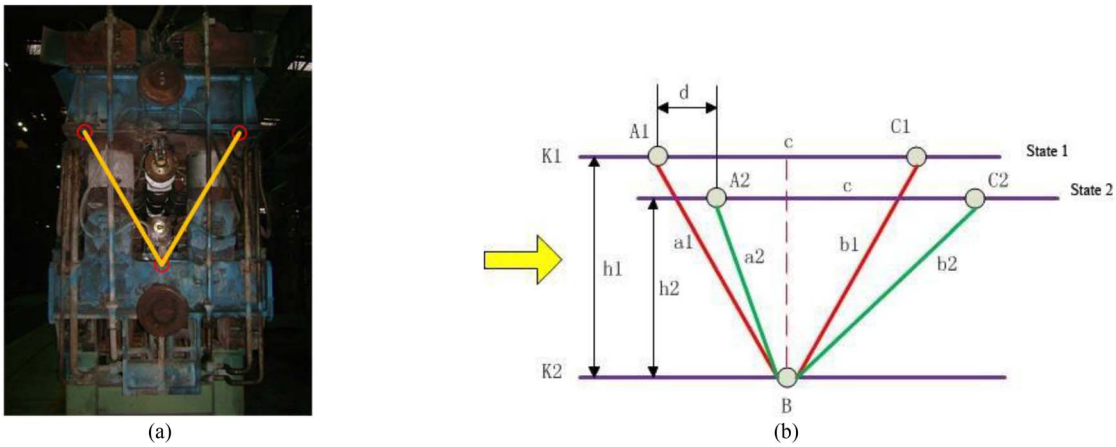

Fig. 8. Working principle of frame displacement. (a) Schematic of sensor installation. (b) Computation method of frame offset/distance.

consistent with the Schema rules, into a standard XML format. If the mapping rules and constraint rules of the data transmitted from the control access layer match the constraints of the well-defined XML Schema, then the data are well-formed. Otherwise, the data are incorrect, and the reasons for the error are recorded and analyzed. Besides, the error logs are recorded for elimination and analysis of failure reasons.

IV. Case Study

In this section, a case study is elaborated to validate the proposed framework using a continuous casting production line in a Chinese steel manufacturing enterprise. The role of the online monitoring system is to obtain the status information of steel casting equipment such as temperature, pressure, and revolving speed.

The system implementation architecture is depicted in Fig. 7. In this paper, status information of equipment, and information of parts and spot inspection collected by a hand-held terminal are transmitted to a remote server through wireless communications. Specifically, data collected by sensors are sent to the ARM center by CAN bus and ZigBee-based wireless network, and field information is inputted into the hand-held terminal by 1-D codes or 2-D codes or RFID. Then, the data are transmitted to the remote server by GPRS and 3G.

The proposed framework has been integrated into the TeamCenter (Siemens PLM software), which is a platform providing MRO services for an industrial machine with a strong expansion capability. The real-time monitoring of operation conditions of the equipment provides some references for further analyses and decisions.

Specifically, if certain observed data are within the normal threshold, it can be concluded that the corresponding parameter is in working order. During the process of continuous steel casting, the stress state of the frame of sector section would cause elastic deformation to the connecting device, which is a critical influencing factor on the quality of the continuous cast slabs. The change of the stress can be monitored by detecting the frame displacement using a magnetostrictive displacement sensor. Fig. 8(a) shows one side of a sector section frame, where each red circle represents the location of sensor installation. In Fig. 8(b), the computation method of the

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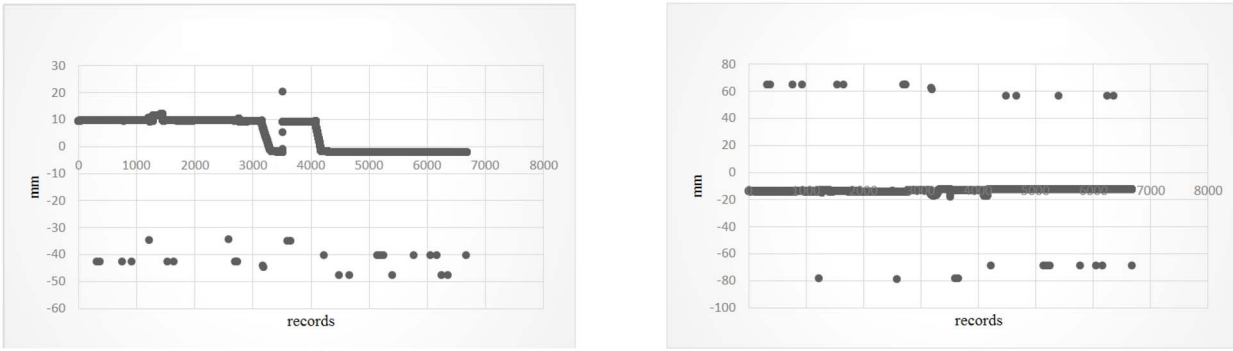

(a) (b)

Fig. 9. Trend charts of frame displacement. (a) Vertical distance. (b) Horizontal offset.

Fig. 10. System user interface.

frame offset/distance is illustrated in details, where K_1 represents the upper part of a frame, and K_2 represents the bottom part. $\overline{BA_1}$ and $\overline{BC_1}$ represent the two waists of the triangle, whose lengths are written as a_1 and b_1. Note that $a_1 = b_1$ in the initial state. A_1 and C_1 denote the initial state, while A_2 and C_2 show the shifted state. The vertical distances between upper part and bottom part of state 1 and state 2 are noted as h_1 and h_2, respectively, and the horizontal offset is noted as d. After the upper frame is shifted, namely in state 2, the length of waists turn to a_2 and b_2. The length of $\overline{A_1C_1}$ is a constant c. We can obtain the value of vertical distances, namely h_1 and h_2, and the horizontal offset d from the MRO system.

State 1: h_1 can be calculated as follows:

$$h_1 = \sqrt{a_1^2 - \left(\frac{c}{2}\right)^2} \quad d = 0. \tag{2}$$

State 2: h_2 can be inferred by

$$\sqrt{p(p-a_2)(p-b_2)(p-c)} = \frac{c \cdot h_2}{2} \quad d = \frac{c}{2} - \sqrt{a_2^2 - h_2^2} \tag{3}$$

where the initial value of c and a_1 are 1350 and 1270 mm, respectively ($\pm$ represents the direction of the offset).

Accordingly, both vertical distances and horizontal offset can be obtained. Their trend charts are illustrated in Fig. 9, where the ordinate axis represents offset/distance and the horizontal axis denotes the data of nearly 7000 records. The data are collected, at a time interval ranging approximately from 10 to 15 s, from a continuous steel casting production line of a certain day. The threshold of normal distance/offset ranges from -100 to 100 mm, which is defined according to experiences. We notice in Fig. 9 that there are few isolated points out of the trend line. However, they are within the threshold.

In addition, the working status and failure information of the equipment can be inquired through the system user interface (Fig. 10). Four parameters, namely, cylinder pressure, bearing temperature, roller speed, and frame displacement, were monitored in this paper. It can be seen from Fig. 10 that the cylinder pressure and framework displacement are in good working condition, while the bearing temperature and the roller speed are abnormal. On the other hand, the bearing temperature

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sensor list shows the sensor ID, value, working state, and acquisition time of each sensor node.

V. Conclusion

Online monitoring in manufacturing industry is crucial to provide fresh and accurate data of environmental conditions and working status of devices for the control center, further to contribute to better MRO services. This paper presents an online monitoring system using IoT-based techniques. The proposed framework introduces an IoT-based monitoring system architecture, and data conversion mechanisms for continuous steel casting production line. The data processing part deals with large amount of heterogeneous data, multiple communication protocols, and provides a data interface. A real-world application case study shows that the system performs well in the real continuous steel casting production environment. The proposed system architecture offers enough flexibility to integrate new hardware and software capabilities. This technology can be well applied to other similar manufacturing environments for real time monitoring. Our future research will focus on applying some techniques of mobile Internet and Big Data analytics to improve the functionality and performance of the proposed system.

References

[1] "Continuous slab casting solutions," Siemens, Accessed on Aug. 10, 2016. [Online]. Available: https://www.yumpu.com/fr/document/view/21875312/

[2] L. E. Redding, C. J. Hockley, R. Roy, and J. Menhen, "The role of maintenance, repair, and overhaul (MRO) knowledge in facilitating service led design: A nozzle guide vane case study," in *9th WCEAM Research Papers* (Lecture Notes Mech. Eng.). Cham, Switzerland: Springer, 2015, pp. 379–395.

[3] J. Pan *et al.*, "An Internet of Things framework for smart energy in buildings: Designs, prototype, and experiments," *IEEE Internet Things J.*, vol. 2, no. 6, pp. 527–537, Dec. 2015.

[4] R. C. A. Alves, L. B. Gabriel, B. T. de Oliveira, C. B. Margi, and F. C. L. dos Santos, "Assisting physical (hydro) therapy with wireless sensors networks," *IEEE Internet Things J.*, vol. 2, no. 2, pp. 113–120, Apr. 2015.

[5] J. A. Stankovic, "Research directions for the Internet of Things," *IEEE Internet Things J.*, vol. 1, no. 1, pp. 3–9, Feb. 2014.

[6] S. Ulukus *et al.*, "Energy harvesting wireless communications: A review of recent advances," *IEEE J. Sel. Areas Commun.*, vol. 33, no. 3, pp. 360–381, Mar. 2015.

[7] C. Pereira and A. Aguiar, "Towards efficient mobile M2M communications: Survey and open challenges," *Sensors*, vol. 14, no. 10, pp. 19582–19608, 2014.

[8] A. Rajandekar and B. Sikdar, "A survey of MAC layer issues and protocols for machine-to-machine communications," *IEEE Internet Things J.*, vol. 2, no. 2, pp. 175–186, Apr. 2015.

[9] P. Marie, T. Desprats, S. Chabridon, M. Sibilla, and C. Taconet, "From ambient sensing to IoT-based context computing: An open framework for end to end QoC management," *Sensors*, vol. 15, no. 6, pp. 14180–14206, 2015.

[10] X. Wei, Q. Li, F. Ye, J. Zhang, and R. Bie, "Building the data association network of sensors in the Internet of Things," *Automatika J. Control Measur. Electron. Comput. Commun.*, vol. 54, no. 4, pp. 459–470, 2013.

[11] M. R. Palattella *et al.*, "On optimal scheduling in duty-cycled industrial IoT applications using IEEE802. 15.4e TSCH," *IEEE Sensors J.*, vol. 13, no. 10, pp. 3655–3666, Oct. 2013.

[12] L. Kong, M. Xia, X.-Y. Liu, M.-Y. Wu, and X. Liu, "Data loss and reconstruction in sensor networks," in *Proc. IEEE INFOCOM*, Turin, Italy, 2014, pp. 1654–1662.

[13] M.-S. Pan, P.-L. Liu, and Y.-P. Lin, "Event data collection in ZigBee tree-based wireless sensor networks," *Comput. Netw.*, vol. 73, pp. 142–153, Nov. 2014.

[14] H. Lu, J. Li, and M. Guizani, "Secure and efficient data transmission for cluster-based wireless sensor networks," *IEEE Trans. Parallel Distrib. Syst.*, vol. 25, no. 3, pp. 750–761, Mar. 2014.

[15] X. Yan, L. Zhang, Y. Wu, Y. Luo, and X. Zhang, "Secure smart grid communications and information integration based on digital watermarking in wireless sensor networks," *Enterprise Inf. Syst.*, pp. 1–27, 2015, doi: 10.1080/17517575.2015.1033767.

[16] F. Ishmanov, S. W. Kim, and S. Y. Nam, "A robust trust establishment scheme for wireless sensor networks," *Sensors*, vol. 15, no. 3, pp. 7040–7061, 2015.

[17] P. Kumar, M. Ylianttila, A. Gurtov, S. G. Lee, and H. J. Lee, "An efficient and adaptive mutual authentication framework for heterogeneous wireless sensor network-based applications," *Sensors*, vol. 14, no. 2, pp. 2732–2755, 2014.

[18] M. Yu, M. Zhou, and W. Su, "A secure routing protocol against byzantine attacks for MANETs in adversarial environments," *IEEE Trans. Veh. Technol.*, vol. 58, no. 1, pp. 449–460, Jan. 2009.

[19] K. Islam, W. Shen, and X. Wang, "Wireless sensor network reliability and security in factory automation: A survey," *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.*, vol. 42, no. 6, pp. 1243–1256, Nov. 2012.

[20] F. Tao, Y. Zuo, L. D. Xu, and L. Zhang, "IoT-based intelligent perception and access of manufacturing resource toward cloud manufacturing," *IEEE Trans. Ind. Informat.*, vol. 10, no. 2, pp. 1547–1557, May 2014.

[21] L. Lu, H. Dong, C. Yang, and L. Wan, "A novel mass data processing framework based on Hadoop for electrical power monitoring system," in *Proc. Asia–Pacific APPEEC*, Shanghai, China, Mar. 2012, pp. 1–4.

[22] P. Gonizzi, G. Ferrari, V. Gay, and J. Leguay, "Data dissemination scheme for distributed storage for IoT observation systems at large scale," *Inf. Fusion*, vol. 22, pp. 16–25, Mar. 2015.

[23] C. Liu, C. Yang, X. Zhang, and J. Chen, "External integrity verification for outsourced big data in cloud and IoT: A big picture," *Future Gener. Comput. Syst.*, vol. 49, pp. 58–67, Aug. 2015.

[24] P. Koedrith *et al.*, "Recent trends in rapid environmental monitoring of pathogens and toxicants: Potential of nanoparticle-based biosensor and applications," *Sci. World J.*, vol. 2015, 12 pp, 2015. [Online]. Available: http://dx.doi.org/10.1155/2015/510982

[25] Y. E. Aslan, I. Korpeoglu, and Ö. Ulusoy, "A framework for use of wireless sensor networks in forest fire detection and monitoring," *Comput. Environ. Urban Syst.*, vol. 36, no. 6, pp. 614–625, 2012.

[26] P. Zhou, G. Huang, L. Zhang, and K. F. Tsang, "Wireless sensor network based monitoring system for a large-scale indoor space: Data process and supply air allocation optimization," *Energy Build.*, vol. 103, pp. 365–374, Sep. 2015.

[27] W. He, K. Li, Q. Zhou, and S. Li, "A CR spectrum allocation algorithm in smart grid wireless sensor network," *Algorithms*, vol. 7, no. 4, pp. 510–522, 2014.

[28] G. Fortino, A. Guerrieri, G. M. P. O'Hare, and A. Ruzzelli, "A flexible building management framework based on wireless sensor and actuator networks," *J. Netw. Comput. Appl.*, vol. 35, no. 6, pp. 1934–1952, 2012.

[29] M.-F. Sriti and P. Boutinaud, "PLMXQuery: Towards a standard PLM querying approach," in *Product Lifecycle Management. Towards Knowledge-Rich Enterprises*. Heidelberg, Germany: Springer, 2012, pp. 379–388.

[30] E. J. Umble, R. R. Haft, and M. M. Umble, "Enterprise resource planning: Implementation procedures and critical success factors," *Eur. J. Oper. Res.*, vol. 146, no. 2, pp. 241–257, 2003.

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